Source: TTI

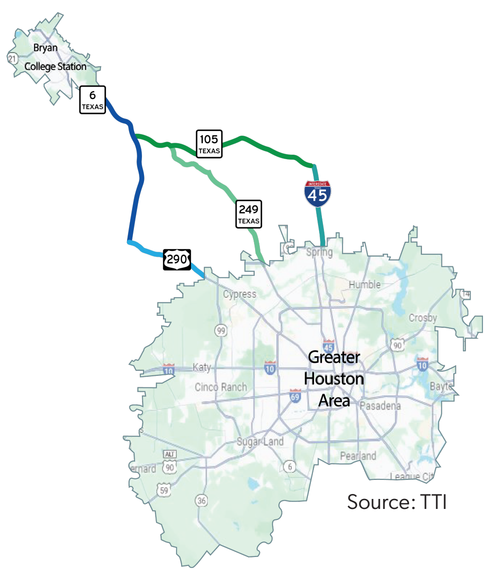
Texas A&M Transportation Institute
Transforming Roads: Unleashing Smart Technologies (TRUST)

In 2024, the United States Department of Transportation (USDOT) awarded cooperative agreements collectively worth $60 million to Maricopa County DOT, Texas A&M Transportation Institute (TTI), and Utah DOT to deploy, operate, and showcase secure, interoperable vehicle-to-everything (V2X) deployments.

Problem

There is significant traffic along the corridors connecting Houston and Bryan/College Station (B/CS) (e.g., US-290, TX-6, TX-249, TX-105 and I-45). Both regions attract over 100,000 eventgoers on busy weekends, and B/CS is home to Texas's largest university, Texas A&M University (TAMU).

Serving more than five million people, these corridors accommodate hundreds of thousands of daily trips along 300 miles of the National Highway System (NHS), creating growing demands for improved safety, coordination, and communication.



Texas TRUST Deployment Area

Approach

TTI is leading the TRUST project that is deploying V2X technology in the mega-region of Texas encompassing the Greater Houston metropolitan area, Bryan/College Station, and connecting routes. The deployment area will leverage existing roadside unit (RSU) deployments that broadcast signal phase and timing (SPaT) messages at over 1,000 intersections in Houston.

An additional 35 intersections near the university in College Station will be equipped with V2X technology, while 50 intersections in the Greater Houston metropolitan area will be upgraded to broadcast SPaT messages. The deployment will also equip 129 fleet vehicles with V2X technology, including transit and emergency response vehicles.

On NHS routes, low-cost, lightweight computing devices will be installed providing capabilities to gather real-time information without the overwhelming infrastructure. The project will use a hybrid communication approach that combines direct V2X and network V2X to provide complementary and redundant data exchange for enhanced safety and mobility.

Source: TTI

City	Use Cases Being Implemented
City of Bryan/College Station	SPaT-Enabled Intersections for Road User Identification and Protection
	Traffic Signal Preemption and Priority
	Transit Fleet Integration
	Every Day a Gameday
	Enhanced Highway Construction Worker Safety
US 290/SH 6/SH 249 Rural Corridor	Incident Management / Hurricane Evacuation
	Adverse Weather Events/ Flooding
	Planned Construction and Special Events
	Curve Speed Warning
Greater Houston Metropolitan Area	Roadway Flood Warning
	SPaT-Enabled Intersections
	Right Turns on Red Warning
	Red Light Violation Warning
	Wrong Way Driving
	Emergency Vehicle Response Time and Safety

Partners

TTI has partnered with proven entities with significant deployment expertise across all aspects of transportation, communications, infrastructure, and vehicle arenas. Partners include Texas A&M University, Texas Department of Transportation, Bryan/College Station Metropolitan Planning Organization (MPO), City of Houston, Missouri City, Fort Bend County, City of College Station, TAMU Internet2 Technology Evaluation Center, Qualcomm, Applied Information, AT&T, Mixon-Hill, AstraZero, Bush Combat Center, General Motors, Audi, Google, Paradigm Traffic Systems, and Integrity Security Systems.

Value Proposition

Texas TRUST aims to deliver measurable safety and mobility improvements across dynamic regions and high-demand corridors. As a result, the TRUST project seeks to accomplish economic development, demonstrate regional benefits, and create transferable and scalable V2X solutions.

For more Information:

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